

Assessment of Heavy Metal Contamination and Pollution Indices in Groundwater from Basement Aquifers of Kibiya, Northwestern Nigeria

^{1,2}Ibrahim Ishaq, ²Musa Bala Gire

¹Department of Geological Sciences, Faculty of Science, Federal University Gusau, PMB 1001, Zamfara State, Nigeria.

²Department of Geology, Faculty of Earth and Environmental Sciences, Bayero University Kano, PMB 3011, Kano State, Nigeria.

*Corresponding authors' email: mbgirei.geo@buk.edu.ng

ABSTRACT

Groundwater is the principal domestic and agricultural water supply in Kibiya town, northwestern Nigeria. However, heavy metal contamination threatens its quality and public health. This study assessed the concentrations, spatial distribution, and pollution status of selected heavy metals in groundwater from Basement Complex aquifers. Thirty samples (ten boreholes, twenty hand-dug wells) were analyzed for Fe, Mn, Zn, Cu, Pb, Cd, and Cr using Microwave Plasma-Atomic Emission Spectrometry (MP-AES). Pollution was evaluated using the Heavy Metal Pollution Index (HPI), Heavy Metal Evaluation Index (HEI), and Contamination Degree (Cd). The results revealed that Pb and Cd are the dominant contaminants. Pb concentrations (0.002-0.095 mg/L) exceeded WHO and NSDWQ drinking-water limits in 83.3% of samples, while Cd (0.002-0.020 mg/L) exceeded limits in 63.3%. The indices yielded contrasting classifications due to differing mathematical sensitivities. The HPI categorized 77% of samples as low pollution, as clean parameters diluted the heavy metal spikes. Conversely, the cumulative Cd index flagged 86.6% as highly contaminated, capturing the unmitigated additive effect of concurrent Pb and Cd exceedances. The HEI presented a balanced middle ground, distributing samples across low (40%), medium (36.7%), and high (23.3%) categories. The study demonstrates that local groundwater quality condition is concerning due to high level of Pb and Cd contamination that poses significant public health threats. Continuous monitoring, improved environmental management, and targeted water treatment are recommended to ensure sustainable groundwater utilization in the area.

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INTRODUCTION

Groundwater is a vital source of freshwater for domestic, agricultural, and industrial uses, particularly in developing countries where it serves as the primary source of potable water. However, its quality can be adversely affected by both natural and anthropogenic processes. In particular, heavy metals are introduced into groundwater through the weathering of rocks and minerals, as well as through activities such as agriculture, industrial operations, waste disposal, and domestic wastewater discharge. Because of their persistence, non-biodegradable nature, and long biological half-lives, heavy metals can accumulate in the environment and pose significant risks to human health (Muchuweti et al., 2006). Elements such as chromium (Cr), nickel (Ni), cadmium (Cd), arsenic (As), lead (Pb), and manganese (Mn) are known to cause a range of health problems, including cardiovascular, neurological, kidney, and bone diseases, as well as carcinogenic, mutagenic, and teratogenic effects (Nagajyoti et al., 2010; Jarup, 2003; Itanna, 2002). Heavy metals occur in groundwater in dissolved, particulate, and colloidal forms and may originate from natural processes such as rock weathering and ore mineralization or from anthropogenic sources including industrial effluents, domestic waste, and agricultural activities (Adepoju-Bello et al., 2009; Marcovecchio et al., 2007).

In crystalline basement terrains, groundwater typically occurs within weathered regolith and fractured crystalline rock units, which control groundwater storage and movement (Clark,

1985; Bala & Eke, 2001; Hamidu et al., 2016). The chemical composition of groundwater in such environments is largely governed by geochemical processes including rock-water interaction, mineral dissolution, and weathering reactions occurring in the subsurface (Adekunle et al., 2007). These natural processes contribute to the enrichment of dissolved ions in groundwater. In addition, anthropogenic activities such as the application of fertilizers and agrochemicals, improper waste disposal, industrial effluents, and sewage infiltration can significantly alter groundwater chemistry by introducing contaminants that may pose risks to human health (Igusi et al., 2001).

Assessment of heavy metal contamination in groundwater commonly involves the application of water quality and pollution indices, which provide an effective means of evaluating contamination levels and comparing them with established drinking-water standards. Several indices, including the Heavy Metal Pollution Index (HPI), Heavy Metal Evaluation Index (HEI), Metal Index (MI), Ecological Risk Index (ERI), and Hazard Quotient (HQ), have been widely used in groundwater quality studies, while also evaluating groundwater quality by comparing measured heavy metal concentrations with permissible limits recommended by national and international water quality standards (Edet and Offiong, 2002; Kwaya et al., 2019; Mgbenu and Egbueri, 2019; Hamidu et al., 2021; Grema et al., 2022). In certain instances, the use of pollution indices have been of great help in the interpretation of complex data matrices for a better

understanding of water quality and a variety of environmental factors and geochemical processes involved in the precipitation of heavy metals into groundwater.

The present study focuses on Kibiya Town and its environs in Kibiya Local Government Area of Kano State, northwestern Nigeria, located between longitudes 8°39'26.31"E and 8°44'45.00"E and latitudes 11°23'21.00"N and 11°31'45.09"N, covering approximately 404 km². The area has an estimated population of 138,618 people (NPCN-KN, 2006) and relies heavily on groundwater for domestic water supply. Previous studies in the area have largely concentrated on groundwater potential and suitability for drinking and irrigation purposes (Clark, 1985; Bala and Eke, 2001; Hamidu et al., 2016), while investigations elsewhere in Kano State have reported elevated concentrations of heavy metals in groundwater (Amoo et al., 2018; Hamidu et al., 2021, Abdullahi et al., 2025). Within Kibiya Local Government Area, Lawal and Idris (2025) delineated weathered and fractured basement aquifers using electrical resistivity surveys. However, their study focused on aquifer characterization and did not evaluate groundwater quality or contamination status. Therefore, this study aims to assess the concentration levels and pollution status of heavy metals in groundwater from Kibiya Town and its environs using selected heavy metal pollution indices, with a view to providing information relevant to groundwater management and public health protection.

Geological Setting and Hydrogeology

Geological Setting

Nigeria is underlain by three major geological provinces, which comprise the Precambrian Basement Complex, the Jurassic Younger Granite Province, and the sedimentary basins of Cretaceous age. The Basement Complex, which occupies about half of the country, consists predominantly of crystalline igneous and metamorphic rocks of Precambrian age and forms part of the Pan-African mobile belt situated between the West African and Congo cratons. It comprises migmatite-gneiss complexes, schist belts, and older granites

emplaced during the Pan-African Orogeny. The Younger Granite Province of north-central Nigeria consists of anorogenic ring complexes emplaced during the Jurassic period, while the sedimentary basins include the Chad Basin, Sokoto Basin, Benue Trough, Niger Delta, Dahomey Basin, and Bida Basin, which contain sediments ranging in age from Cretaceous to Recent.

The geology of Kano State reflects the major geological units present in northern Nigeria and can be broadly divided into three lithological groups. The most extensive unit is the Precambrian Basement Complex, which underlies more than 70% of the state and consists mainly of crystalline igneous and metamorphic rocks, predominantly granites and gneisses. These rocks constitute the principal aquifer host in the area, with groundwater occurrence largely restricted to weathered and fractured zones. The second geological unit comprises the Jurassic Younger Granite rocks, which intruded the Precambrian Basement Complex and occur mainly around Ririwai in Doguwa Local Government Area. These intrusive bodies form part of the Nigerian Younger Granite Province and are associated with ring-complex structures and mineralization. The third unit is represented by the Quaternary sedimentary deposits of the Chad Formation, composed predominantly of sandstone and unconsolidated sediments that unconformably overlie the Basement Complex rocks. These sediments are restricted to the northeastern parts of Kano State, particularly around Danbatta and its environs (MacDonald and Partners, 1986).

The study area, Kibiya and its environs, is located within the Basement Complex terrain of Kano State. It is underlain predominantly by Precambrian crystalline rocks (Figure 1). The geology is characterized largely by porphyritic biotite granites that have undergone varying degrees of weathering and fracturing, creating secondary porosity and permeability that control groundwater occurrence and movement within the area. These geological conditions play a significant role in influencing the hydrogeochemistry and heavy metal content of groundwater in the study area.

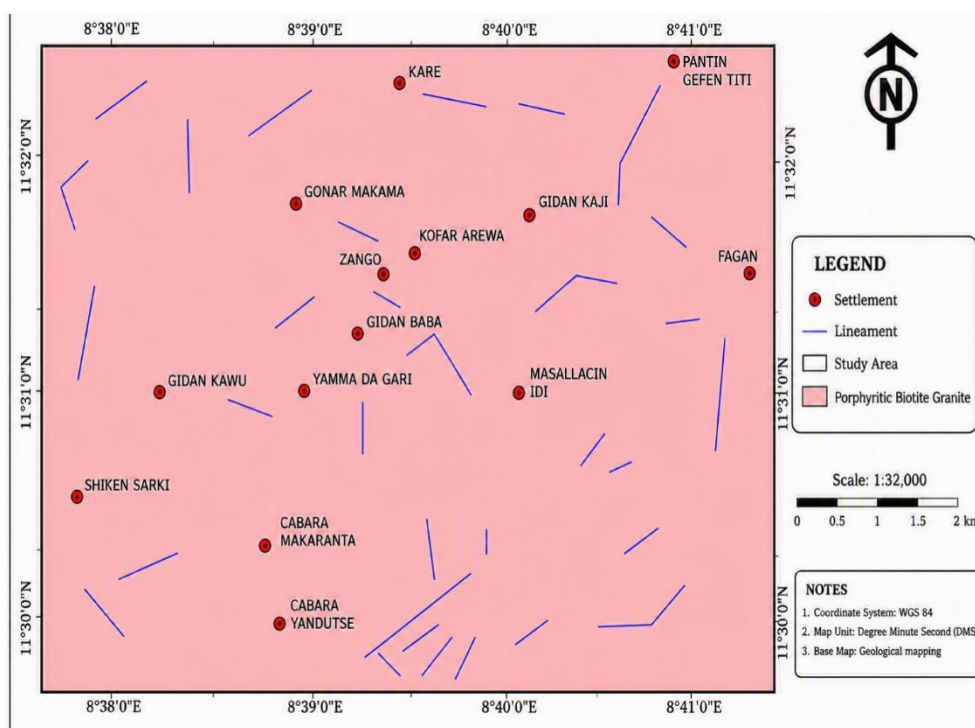


Figure 1: Geologic Map of the Study Area

Hydrogeology

Groundwater occurrence in Kano State is controlled primarily by lithology, weathering, and structural features. In the Basement Complex terrain, groundwater is stored mainly within the weathered regolith and fractured bedrock, whereas the Chad Formation contains porous sedimentary aquifers. The thickness of the weathered zone, fracture density, and degree of weathering largely determine aquifer productivity (Muhammad, 1984; Alagbe, 1987; Uma & Kehinde, 1994).

Previous studies have shown that the weathered regolith forms the principal aquifer in most parts of the Kano Basement Complex, while fractures enhance groundwater storage and transmission (Bala et al., 2011). Consequently, groundwater occurrence and yield vary considerably across the state depending on the local geological and hydrogeological conditions. The hydrogeological map of the study area is presented in Figure 2.

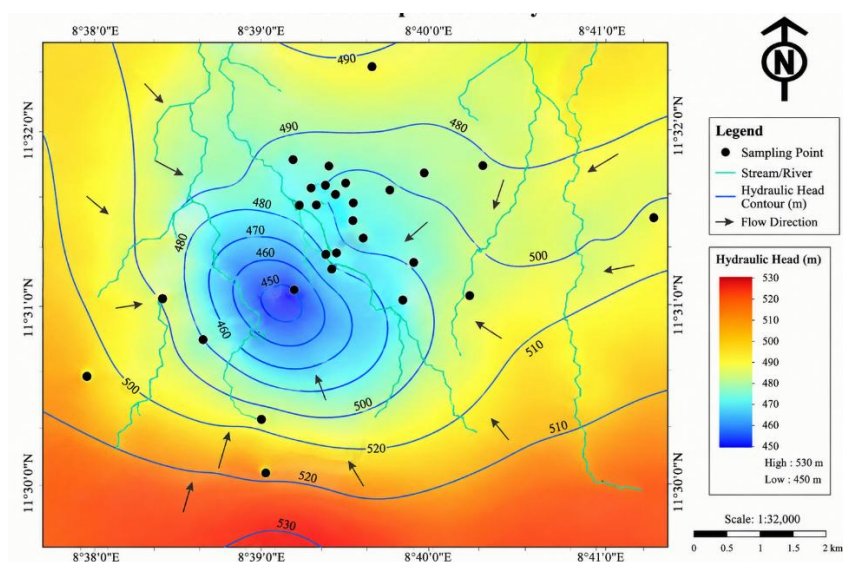


Figure 2: Hydrogeological Map of Groundwater Flow Direction in the Study Area

MATERIALS AND METHODS

Groundwater Sampling

Groundwater samples totaling 30 in number comprising of twenty hand pumps and ten dug wells across the project area were representatively collected in Pre-cleaned 120 mL polyethylene bottles. The sampling was conducted in the month of June, 2026 during the wet season. In order to obtain the correct samples from each sampling station the water sources were bailed or purge out for 10 minutes to eliminate water from casing storage in boreholes and dug wells so that fresh uncontaminated water coming directly from the aquifers were sampled. The coordinates and the elevations at each sampling point were measured with the GPS while the physical parameters of TDS and pH of water were measured in-situ at each location. The water levels or depth to groundwater levels in wells were measured using dip meter/measuring tape. Few drops of concentrated nitric acid were added to each of the groundwater sample, this was done to slow down or completely stop any possible metal precipitation out of solution and to further lower the water pH to 2.0, this will also stop any chemical reactions that might occur in groundwater which can alter the real results expected from the analysis of the samples. The samples were appropriately labelled covered tightly making sure no air was trapped inside the samples, stored in ice-packed cooling container and transported to the laboratory for further preparation and analysis. The APHA (2022) procedure were followed to during the sampling.

Heavy Metal Analysis

Sample Digestion/Analysis

Laboratory analyses were conducted to determine the heavy metal concentrations of the groundwater samples. Physicochemical parameters measured included pH, electrical

conductivity (EC), and total dissolved solids (TDS). The heavy metals (Fe, Mn, Zn, Cu, Pb, Cd and Cr) were determined using Microwave Plasma–Atomic Emission Spectrometry (MP-AES) following APHA (2017) standard procedures. All analyses were carried out at the World Health Organization/Federal Ministry of Water Resources Laboratory, Kano. The MP-AES consists of a microwave-induced plasma interfaced to an Atomic Emission Spectrophotometer (AES). It is used for simultaneous multi-analyte determination of major and minor elements. MP-AES employs microwave energy to produce a plasma discharge using Nitrogen supplied from a gas cylinder or extracted from ambient air, which eliminates the need for sourcing gases in remote locations or foreign countries. Samples were nebulized before interaction with the plasma in MP-AES measurements. The atomized sample passes through the plasma and electrons are promoted to the excited state. The light emitted electrons return to the ground state, the light is separated into a spectrum and the intensity of each emission line is measured at the detector. Most commonly determined elements can be measured with a working range of low part per million (ppm) to weight percent (wt. %). MP-AES is a technique comparable to traditional AA and AES but with several potential advantages including lower cost of operation and elimination of the requirement for flammable gasses.

Location and Spatial Distribution Maps and Statistical Evaluations

The location and Spatial maps of the study were produced using the Arc GIS software 10.5 version to interpolate, classified and reclassified the data and produced the required maps in each case. In order to evaluate the pollution status of each groundwater collected from the study area in terms of their heavy metal content, different pollution indices were

used for the evaluation of these includes, contaminant degree (C_d) heavy metals pollution index (HPI), and heavy metals evaluation index (HEI).

Heavy Metal Pollution Index (HPI)

Mohan *et al.*, (1996) proposed the weighted arithmetic quality mean method which involved two steps. One is to develop rating scale for each of the selected parameters and a weight (W_i). The second step involved selection of the pollution parameter which the index is to be based on. The rating is any value between zero and one which depends on its importance on the quality consideration. This was obtained by making value inversely proportional to the recommended standard (S_i) for corresponding parameter.

The HPI is calculated using the formula

$$\frac{\sum_{i=1}^n W_i Q_i}{\sum_{i=1}^n W_i}$$

Where Q_i the sub-index of i th parameter, W_i the unit weightage of the i th parameter, n is the number of parameters considered

The Q_i sub-index is computed using the formula:

$$Q_i = \sum_{i=1}^n \frac{\{Mi(-)Ii\}}{(S_i - I_i)} \times 100$$

Mi I_i and S_i were the Heavy metal's i th parameter monitored, ideal and standard values respectively. The negative sign (-) is the numerical difference of the two values, the algebraic sign is ignored for this index. HPI was calculated using Pb, Cd, Cr, Fe, Zn, Cu, and Mn, with unit weights inversely proportional to their permissible limits. An HPI value of 100 is commonly treated as a critical threshold: values below 100 indicate lower aggregate metal pollution, whereas values above 100 indicate a high pollution load (Mohan *et al.*, 1996).

Heavy Metal Evaluation Index (HEI)

The HEI provide the overall quality of water as regard the heavy metals detected in the groundwater; the equation used for computing this index is

$$HEI = \sum_{i=1}^n \frac{H_c}{H_{mac}}$$

Where H_c is the monitored value of the i th parameter and H_{mac} is the minimum admissible concentration of the i th parameter. The HEI threshold is proposed in this study (Table 4).

Contamination Index (C_d)

This index takes into account both the number of parameters exceeding the upper permissible limits of the potentially harmful elements, and the concentrations exceeding these limit values (Backman *et al* 1998). Contamination degree, C_d , is calculated for each of the groundwater sample analyzed, this equal to the sum of the contamination factors of individual parameter exceeding the upper permissible. To calculation the C_d the following steps and equations were used. A contamination degree of greater than 3 is treated is high contamination, while a threshold of between 1 to 2 is referred to as medium contamination, where as a value of less than 1 is treated as low contamination (Edet and Offiong, 2002).

$$C_d = \sum_{i=1}^n Cfi$$

To get Cfi we used $Cfi = \frac{CAi}{CNI} - 1$

Cfi is the contamination factor for the i -th component,

CAi is analytical value of the i -th component,

CNI is upper permissible concentration of the i -th component

(N stand for the normative value).

RESULTS AND DISCUSSION

The concentrations of heavy metals in groundwater from Kibiya and its environs are presented in **Table 1**, while their statistical summary is shown in **Table 2**. Spatial distribution patterns of the analyzed metals are illustrated in **Figures 3a–3g**. The results reveal varying degrees of metal enrichment across the study area, reflecting the combined influence of geological and anthropogenic factors.

Heavy Metals Distribution in Groundwater

Iron (Fe^{3+})

Iron concentrations range from **0.11 to 0.61 mg/L**, with a mean value of **0.24 mg/L**. Although the average concentration is below the WHO (2011) and NSDWQ (2017) guideline value of 0.30 mg/L, eight groundwater samples (S002, S003, S008, S009, S022, S023, S024, and S025) exceeded the permissible limit. The elevated iron concentrations are most likely related to the weathering of iron-bearing minerals, particularly biotite and accessory ferromagnesian minerals present in the porphyritic biotite granite. Local reducing conditions within the aquifer may also enhance iron dissolution. Although iron is not generally considered a major health concern at the observed concentrations, elevated levels may impart an unpleasant taste, stain plumbing fixtures and laundry, and promote bacterial growth in water distribution systems. The localized exceedances therefore indicate areas where groundwater treatment may be required before domestic use.

Lead (Pb)

Lead concentrations range from **0.002 to 0.095 mg/L**, with a mean value exceeding the WHO (2011) and NSDWQ (2017) guideline value of 0.01 mg/L. Notably, 25 of the 30 groundwater samples (83.3%) exceed the recommended drinking water limit, making lead one of the principal contaminants identified in the study area.

The widespread occurrence of elevated lead concentrations suggests contributions from both geogenic and anthropogenic sources. Geogenic inputs may result from the weathering of trace lead-bearing minerals within the granitic basement, whereas anthropogenic contributions may arise from agricultural chemicals, waste disposal practices, atmospheric deposition, and corrosion of plumbing materials. Chronic exposure to elevated lead concentrations has been associated with neurological disorders, impaired cognitive development in children, kidney dysfunction, and cardiovascular diseases. The high frequency of lead exceedance indicates that groundwater from affected locations should not be consumed without appropriate treatment and continuous monitoring.

Copper (Cu)

Copper concentrations vary between **0.21 and 0.47 mg/L**, with an average value of **0.336 mg/L**, and all samples are below the WHO (2011) and NSDWQ (2017) guideline values. Copper is commonly derived from the weathering of copper-bearing minerals and, in some cases, from corrosion of metallic plumbing materials. The low concentrations recorded indicate that copper does not constitute a groundwater quality concern within the study area and is unlikely to pose adverse health effects.

Cadmium (Cd)

Cadmium concentrations range from **0.002 to 0.020 mg/L**, with 19 of the 30 groundwater samples exceeding the WHO (2011) and NSDWQ (2017) guideline value of 0.003 mg/L. The elevated cadmium concentrations indicate localized groundwater contamination that may be associated with

agricultural inputs, phosphate fertilizers, waste disposal, and natural weathering of trace cadmium-bearing minerals. Cadmium is a highly toxic element that accumulates in the human body, particularly in the kidneys, and prolonged exposure may result in renal impairment, skeletal disorders, and other chronic health effects. Consequently, groundwater containing elevated cadmium concentrations should be treated before domestic consumption.

Manganese (Mn)

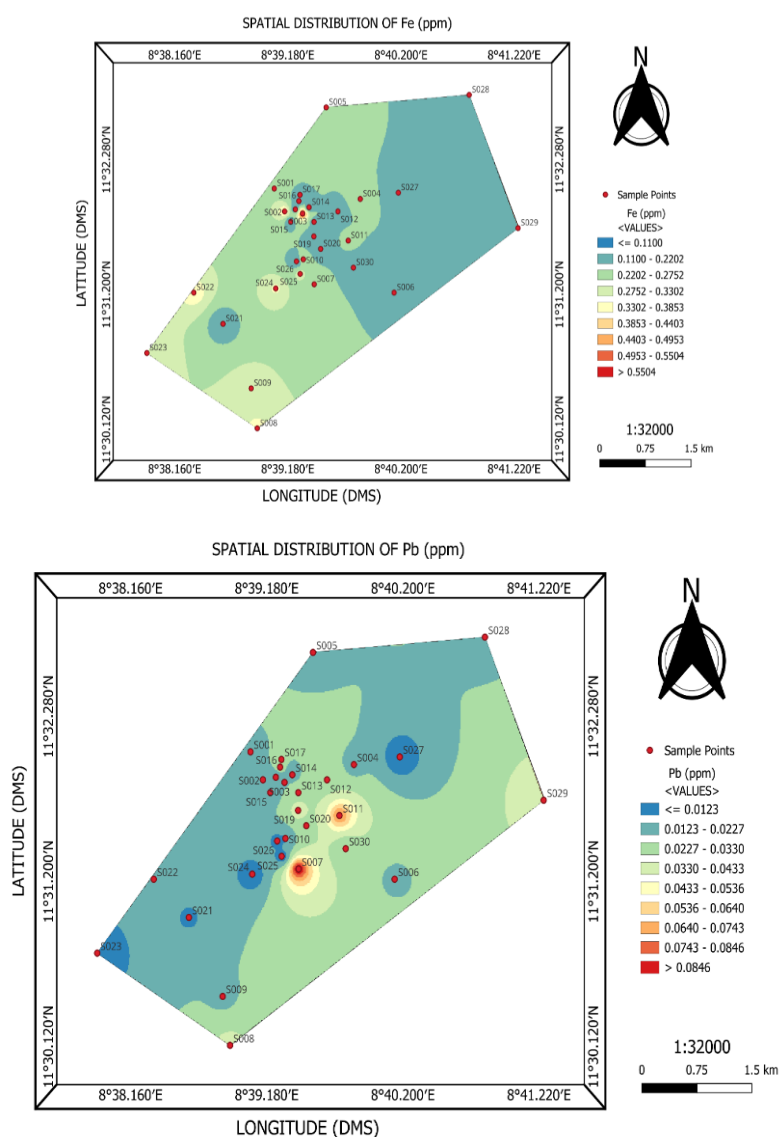
Manganese concentrations range from **0.068 to 0.225 mg/L**, with a mean value of **0.125 mg/L**, and all samples comply with the WHO (2011) and NSDWQ (2017) drinking water standards. The relatively low manganese concentrations are consistent with the weathering of ferromagnesian minerals within the granitic aquifer. The results indicate that manganese is not a significant groundwater quality concern in the study area.

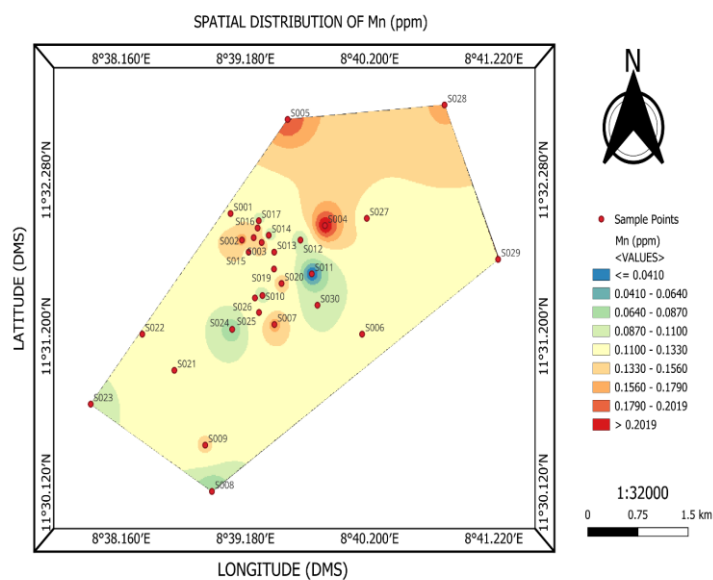
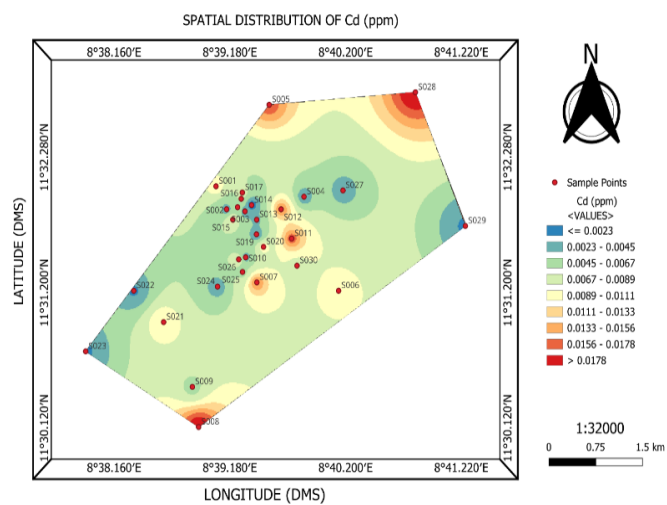
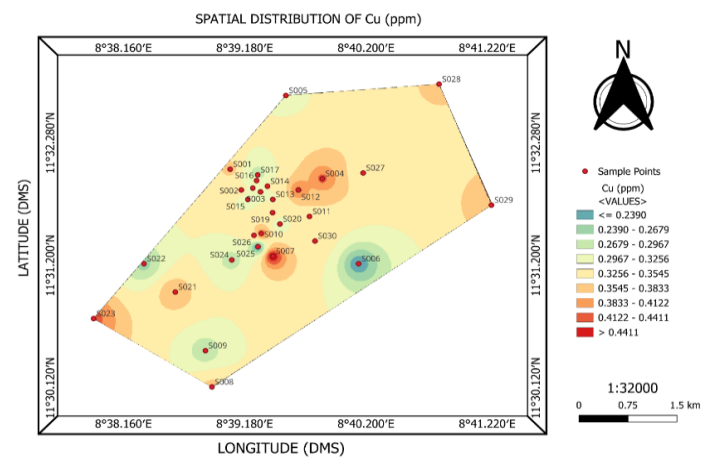
Chromium (Cr)

Chromium concentrations range from **0.01 to 0.06 mg/L**, with an average value of **0.03 mg/L**. Although most groundwater samples comply with the drinking water standards, five samples (S003, S017, S021, S025 and S028) exceed the guideline value of 0.05 mg/L. The localized chromium enrichment may be related to the weathering of chromium-bearing accessory minerals or anthropogenic inputs associated with agricultural activities. The elevated concentrations recorded in a few locations warrant periodic monitoring because long-term exposure to high chromium concentrations may present health risks.

Zinc (Zn)

Zinc concentrations range from **0.11 to 0.68 mg/L**, with an average value of **0.25 mg/L**, and all groundwater samples comply with the WHO (2011) and NSDWQ (2017) drinking water standards. The low zinc concentrations indicate minimal contribution from both geogenic and anthropogenic sources. Zinc is therefore not considered a limiting factor affecting groundwater quality in the study area.





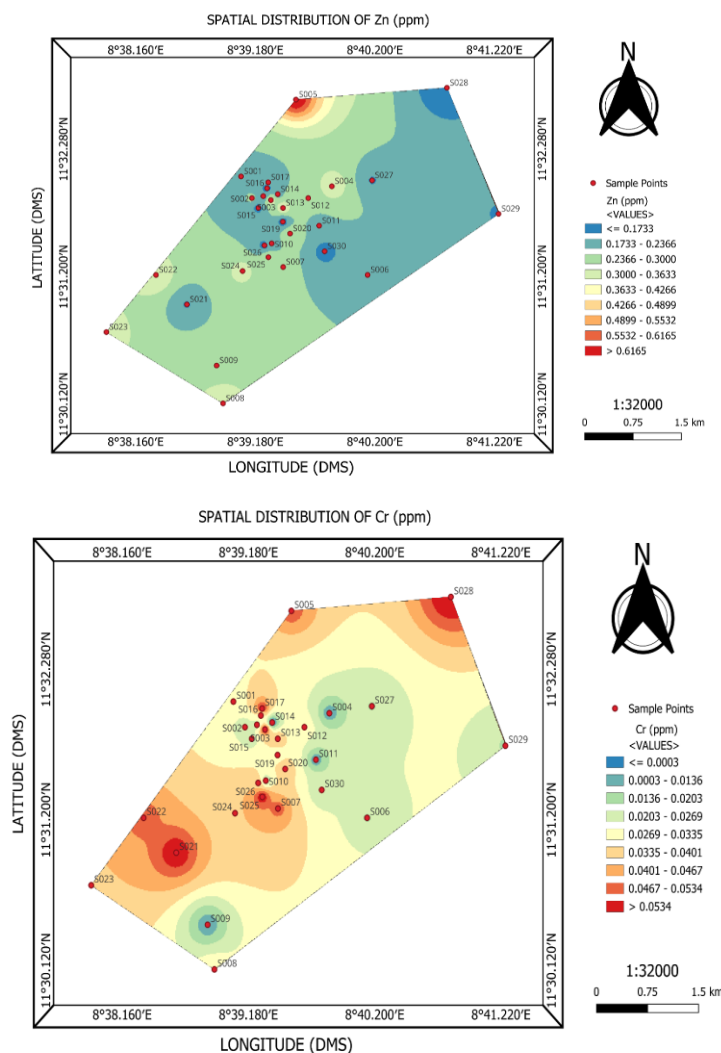


Figure 3: 3a-3g: Spatial Concentration Maps of Fe, Pb, Cu, Cd, Mn, Cr and Zn from the Study Area

Table 1: Results of the Analysis of Heavy Metals Concentrations in Groundwater of the Study Area

Sampl e ID	HDW/ HPB	Locations	Northing Resding Degrees	Easting Reading Degrees	Pb (mg/l)	Cd (mg/l)	Cr (mg/l)	Fe (mg/L)	Zn (mg/L)	Cu (mg/l)	Mn (mg/l)
S001	HDW	GONAR MAKAMA	11° 32' 03.8"	8° 39' 03.5"	0.021	0.011	0.03	0.27	0.21	0.37	0.113
S002	HDW	YAN TIPA	11° 31' 52.4"	8° 39' 09.1"	0.022	0.003	0.02	0.41	0.26	0.34	0.171
S003	HDW	MASALLACIN TAMIYAWA	11° 31' 51.1"	8° 39' 18.8"	0.022	0.003	0.06	0.61	0.37	0.31	0.161
S004	HDW	KWANAR NARIYA	11° 31' 58.8"	8° 39' 50"	0.018	0.003	0.01	0.27	0.37	0.42	0.225
S005	HDW	KARE	11° 32' 42"	8° 39' 31.6"	0.018	0.017	0.05	0.25	0.68	0.31	0.2
S006	HDW	MASALLACIN IDI	11° 31' 13.5"	8° 40' 8.3"	0.02	0.01	0.02	0.16	0.21	0.22	0.124
S007	HDW	TASHAR RUWA	11° 31' 1705"	8° 39' 25.1"	0.095	0.016	0.05	0.26	0.26	0.47	0.168
S008	HDW	CABARA YANDUTSE	11° 30' 7.8"	8° 38' 54.3"	0.036	0.02	0.03	0.34	0.33	0.36	0.068
S009	HDW	CABARA MAKARANTA	11° 30' 27.6"	8° 38' 51"	0.021	0.006	0.01	0.31	0.29	0.28	0.136
S010	HDW	GIDAN BABA	11° 31' 29"	8° 39' 19.2"	0.011	0.003	0.02	0.28	0.22	0.42	0.094
S011	HPB	MASALLACIN JUMUA BABBA	11° 31' 38.3"	8° 39' 43.5"	0.076	0.017	0.01	0.24	0.21	0.35	0.018
S012	HDW	KOFAR AREWA	11° 31' 52.3"	8° 39' 37.9"	0.028	0.016	0.03	0.18	0.26	0.42	0.096
S013	HDW	ZANGO	11° 31' 47.6"	8° 39' 25"	0.033	0.004	0.04	0.19	0.38	0.32	0.126
S014	HDW	SHAGON ISAU	11° 31' 54.8"	8° 39' 22.3"	0.017	0	0	0.14	0.21	0.33	0.081
S015	HDW	KAN KWANA	11° 31' 47.2"	8° 39' 12.4"	0.014	0.01	0.02	0.14	0.14	0.31	0.123
S016	HPB	GIDAN DAN DARA	11° 31' 57.5"	8° 39' 16.8"	0.026	0.007	0.03	0.13	0.13	0.34	0.161
S017	HPB	LAYIN ALHAJI DAN KAROTA	11° 32' 00.4"	8° 39' 17.4"	0.023	0.008	0.06	0.21	0.21	0.25	0.084
S018	HPB	BAYAN TASHA	11° 31' 53.9"	8° 39' 14.9"	0.018	0.006	0.02	0.16	0.16	0.31	0.161

Sample ID	HDW/HPB	Locations	Northing Resding Degrees	Easting Reading Degrees	Pb (mg/l)	Cd (mg/l)	Cr (mg/l)	Fe (mg/L)	Zn (mg/L)	Cu (mg/l)	Mn (mg/l)
S019	HPB	GIDAN MALAN ILIYA	11° 31' 40.9"	8° 39' 24.9"	0.041	0.002	0.03	0.13	0.13	0.36	0.126
S020	HPB	KOFAR YAMMA	11° 31' 34.3"	8° 39' 28.6"	0.03	0.01	0.04	0.16	0.31	0.29	0.15
S021	HPB	GINDUWA SHIKE	11° 30' 58.3"	8° 38' 35.8"	0.011	0.011	0.06	0.17	0.17	0.38	0.123
S022	HDW	GIDAN KAWU	11° 31' 13.0"	8° 38' 20"	0.02	0.002	0.05	0.36	0.36	0.26	0.133
S023	HDW	SHIKEN SARKI	11° 30' 44.2"	8° 37' 54.6"	0.002	0.002	0.04	0.33	0.33	0.42	0.096
S024	HDW	YAMMA DA GARI	11° 31' 15.8"	8° 39' 4.3"	0.003	0.003	0.04	0.34	0.34	0.28	0.068
S025	HPB	ITQAAN SCHOOL	11° 31' 22.9"	8° 39' 17.5"	0.006	0.006	0.06	0.3	0.3	0.21	0.126
S026	HPB	GIDAN ALBA	11° 31' 28.2"	8° 39' 15.5"	0.008	0.008	0.04	0.14	0.14	0.34	0.111
S027	HDW	GIDAN KAJI	11° 32' 01.1"	8° 40' 10.6"	0.003	0.003	0.02	0.17	0.17	0.34	0.126
S028	HDW	PANTIN GEFEN TITI	11° 32' 48.9"	8° 40' 48.9"	0.02	0.02	0.06	0.11	0.11	0.36	0.161
S029	HDW	FAGAN	11° 31' 44.6"	8° 41' 15.3"	0.043	0.002	0.02	0.2	0.17	0.37	0.126
S030	HPB	KOFAR KUDU	11° 31' 25.8"	8° 39' 46.3"	0.023	0.008	0.02	0.17	0.13	0.34	0.087

Table 2: Statistical Summary of Heavy Metals Concentration in the Study Area

Parameter	Unit	Minimum	Maximum	Mean	SD
Pb	mg/l	0.002	0.095	0.024	0.020
Cd	mg/l	0.000	0.020	0.008	0.006
Cr	mg/l	0.000	0.060	0.033	0.017
Fe	mg/l	0.110	0.610	0.238	0.107
Zn	mg/l	0.110	0.680	0.252	0.117
Cu	mg/l	0.210	0.470	0.336	0.061
Mn	mg/l	0.018	0.225	0.125	0.042
pH		6.110	9.960	7.646	0.781
TDS	mg/l	106.000	875.000	323.733	167.564
EC	us/cm	115.000	972.000	551.933	242.615
Turbidity	NTU	20.000	94.000	43.800	17.008

Heavy Metal Pollution Assessment

The degree of groundwater contamination by heavy metals was evaluated using the Heavy Metal Pollution Index (HPI), Heavy Metal Evaluation Index (HEI), and Contamination Degree (Cd). The results of both heavy metals analysis and pollution indices evaluations is contained in Table 3 and 4, while their spatial distributions are presented in Figures 4a–4c.

Heavy Metal Pollution Index (HPI)

This index showed that 23 (77 %) out of the analyzed groundwater were of low pollution based on the classification of Mohan et al. (1996). with values below 100, while 7 samples fall into the high pollution class with values above 100, this class constituted 23% of the total water used. However, it is evidently clear from the results obtained that the area is safe with regard to the heavy metal pollution index since majority of the sampled groundwater were within the low pollution class.

Heavy Metal Evaluation Index (HEI)

The values of the HEI were distributed into the three categories of pollution and unpolluted as proposed in this study as <20 low pollution class, 20-40 as medium polluted water and >40 as highly polluted groundwater in the study area. Based on this classification, 12, 11, and 7 groundwater

samples were of low, medium and high groundwater pollution status, which is equivalent to 40, 36.7, and 23.3 % respectively. Generally, it can be seen that majority of the samples fall under low to medium categories of pollution class. However, the exceedance of Pb and Cd in most of the sampling locations calls for an appropriate treatment of the groundwater in the study area before consumption.

Contamination Degree (Cd)

The computed contamination degree (table 3) for the groundwater in the study revealed that 2 samples (6.7%) had a low level of contamination, 2 samples (6.7%) had a medium level, and 26 samples (86.6%) had a high level degree of contamination with respect to their heavy metals' concentrations. The classification used was based on the adopted classification of Edet and Offiong (2002), contained in table 4, low degree class of contamination is less than 1, medium was between 1-3, and high class was more than 3. The variations in pollution classifications across the HPI, HEI, and Cd indices highlight the distinct mathematical sensitivities inherent to each index's structure. In this study, the HPI categorized the majority of samples (77%) as low pollution, whereas the Cd index flagged 86.6% of the samples as highly contaminated. This divergence is attributed to the weighting and averaging mechanisms within the indices. The HPI relies on a weighted averaging system where parameters

conforming to standard limits such as Cu, Mn, and Zn in this study mathematically dilute the heavy impact of toxic outliers like Pb and Cd. Conversely, the Contamination Degree (Cd) operates on a purely cumulative summation framework. By directly adding the individual contamination factors without an averaging denominator, the Cd index prevents clean parameters from masking localized spikes. Because Pb and Cd exceeded permissible drinking water limits in 83.3% and 63.3% of the samples, respectively, their unmitigated additive effect consistently pushed the Cd values past the high

pollution threshold. The HEI offered an intermediate gradient, distributing the samples across low (40%), medium (36.7%), and high (23.3%) categories. This is because HEI assesses the direct ratio of concentrations to admissible limits without applying inverse proportional weighting factors. These results demonstrate that while HPI provides a generalized overview of composite water suitability, the Cd index functions as a highly sensitive environmental indicator that reliably captures cumulative toxic risks posed by multiple reoccurring contaminants.

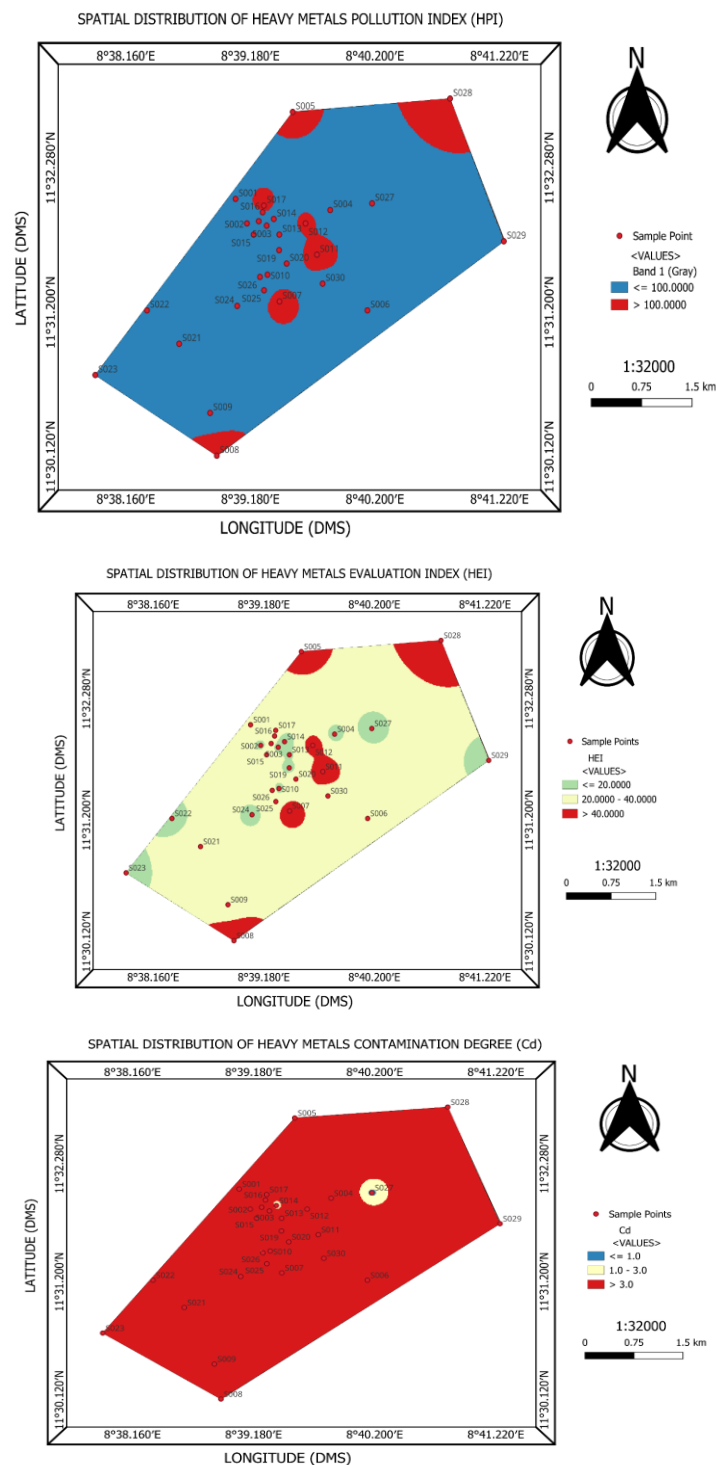


Figure 4: 4a-4c Showing Spatial Distribution Maps of HPI, HEI and Cd Respectively

Table 3: Results of Calculated Pollution Indices of the Study Area

Sample	COMMUNITY	LONGITUDE	LATITUDE	HEI	HPI	Cd
S001	GONAR MAKAMA	11° 32' 03.8"	8° 39' 03.5"	40.54992	92.8	8.649917
S002	YAN TIPA	11° 31' 52.4"	8° 39' 09.1"	14.26608	36.5	10.56608
S003	MASALLACIN TAMIYAWA	11° 31' 51.1"	8° 39' 18.8"	15.75192	37.65	18.05192
S004	KWANAR NARIYA	11° 31' 58.8"	8° 39' 50"	13.28958	33.68	5.389583
S005	KARE	11° 32' 42"	8° 39' 31.6"	60.73167	134.17	10.23167
S006	MASALLACIN IDI	11° 31' 13.5"	8° 40' 8.3"	36.47767	84.72	4.277667
S007	TASHAR RUWA	11° 31' 1705"	8° 39' 25.1"	65.06367	176.29	17.86367
S008	CABARA YANDUTSE	11° 30' 7.8"	8° 38' 54.3"	72.307	166.35	15.507
S009	CABARA MAKARANTA	11° 30' 27.6"	8° 38' 51"	23.604	56.8	7.904
S010	GIDAN BABA	11° 31' 29"	8° 39' 19.2"	12.74017	29.25	5.140167
S011	MASALLACIN JUMUA BABBA	11° 31' 38.3"	8° 39' 43.5"	65.51617	169.95	14.71617
S012	KOFAR AREWA	11° 31' 52.3"	8° 39' 37.9"	57.654	132.73	8.054
S013	ZANGO	11° 31' 47.6"	8° 39' 25"	18.38483	50.93	5.084833
S014	SHAGON ISAU	11° 31' 54.8"	8° 39' 22.3"	2.421917	11.13	-0.37808
S015	KAN KWANA	11° 31' 47.2"	8° 39' 12.4"	35.83242	80.87	3.032417
S016	GIDAN DAN DARA	11° 31' 57.5"	8° 39' 16.8"	27.22025	67.54	3.12025
S017	LAYIN ALHAJI DAN KAROTA	11° 32' 00.4"	8° 39' 17.4"	31.08267	203.84	6.382667
S018	BAYAN TASHA	11° 31' 53.9"	8° 39' 14.9"	22.98192	55.07	2.781917
S019	GIDAN MALAN ILIYA	11° 31' 40.9"	8° 39' 24.9"	12.05483	41.53	2.954833
S020	KOFAR YAMMA	11° 31' 34.3"	8° 39' 28.6"	37.9525	91.68	5.7525
S021	GINDUWA SHIKE	11° 30' 58.3"	8° 38' 35.8"	39.81075	87.11	4.91075
S022	GIDAN KAWU	11° 31' 13.0"	8° 38' 20"	11.14992	28.78	8.949917
S023	SHIKEN SARKI	11° 30' 44.2"	8° 37' 54.6"	9.110667	16.94	6.010667
S024	YAMMA DA GARI	11° 31' 15.8"	8° 39' 4.3"	12.50367	24.64	6.703667
S025	ITQAAN SCHOOL	11° 31' 22.9"	8° 39' 17.5"	23.0365	48.46	7.0365
S026	GIDAN ALBA	11° 31' 28.2"	8° 39' 15.5"	28.97775	63.31	2.17775
S027	GIDAN KAJI	11° 32' 01.1"	8° 40' 10.6"	11.52483	24.1	0.624833
S028	PANTIN GEFEN TITI	11° 32' 48.9"	8° 40' 48.9"	70.49025	167.59	6.79025
S029	FAGAN	11° 31' 44.6"	8° 41' 15.3"	12.3065	42.61	5.3065
S030	KOFAR KUDU	11° 31' 25.8"	8° 39' 46.3"	30.16842	72.38	4.268417

Table 4: Computed Pollution Indices in the Study Area

Pollution Index	Source of classification	Class of pollution	Degree/ level of Pollution	Number of Samples	% of Samples in each class
HPI	Adopted from Mohan et al (1996)	<100	Low	23	77
		>100	High	7	23
HEI	Proposed in this study	<20	Low	12	40
		20-40	Medium	11	36.7
		>40	High	7	23.3
Cd	Adopted from Edet and Offiong (2002)	< 1	Low	2	6.7
		1-3	Medium	2	6.7
		>3	High	26	86.6

Implications for Groundwater Quality

Overall, the groundwater in Kibiya and its environs exhibits relatively low concentrations of Fe, Cu, Mn, Cr and Zn, most of which comply with international and national drinking water standards. However, the widespread occurrence of elevated Pb and Cd concentrations represents a significant water quality concern. All sampled groundwater sources exceed the permissible limits specified by the NSDWQ (2017) and WHO (2022) for drinking water (Table 7), indicating that turbidity constitutes the principal physical

water quality concern within the study area. The elevated turbidity may be attributed to suspended clay particles, weathered rock fragments, organic matter, and poor sanitary conditions around some wells. The shallow nature of several hand-dug wells may also permit the infiltration of fine sediments during rainfall events. Although turbidity does not directly indicate chemical contamination, high turbidity may reduce the aesthetic quality of water and interfere with effective disinfection by providing protective surfaces for microorganisms. Consequently, groundwater intended for

domestic consumption should undergo appropriate treatment before use. The pollution index results indicate that although groundwater quality is generally acceptable in many parts of the study area, localized contamination hotspots exist and may pose risks to public health. These findings underscore the need for routine groundwater quality monitoring, improved waste management practices, regulation of agrochemical use, and appropriate treatment of contaminated water sources before domestic consumption.

CONCLUSION

The groundwater in Kibiya town and its environs exhibits a moderate hydrochemical characteristics that may pose some risk for domestic use in most parts of the study area. Most of the analyzed heavy metals, including Cu, Mn, and Zn, occur within the permissible limits of WHO (2011) and NSDWQ (2017). However, elevated concentrations of Pb, Cd, and Fe, together with high turbidity values at some locations, indicate deterioration of groundwater quality. These variations are attributed to both natural geochemical processes and anthropogenic influences, particularly agricultural activities, poor sanitation, and indiscriminate waste disposal.

The heavy metal pollution indices revealed spatially variable contamination across the study area. The HPI classified 77% of the samples as having low pollution and 23% as highly polluted, while the HEI indicated low, medium, and high pollution classes for 40%, 36.7%, and 23.3% of the samples, respectively. In contrast, the Cd assessment identified high contamination levels in 86.6% of the samples, highlighting the cumulative impact of heavy metals, particularly Pb and Cd. The pollution index classifications vary because the indices measure contamination differently. Consequently, regular groundwater quality monitoring, improved environmental sanitation, and treatment of contaminated water sources are recommended before consumption to protect public health and ensure the sustainable utilization of groundwater resources in the area. Additionally, further investigation is required to establish contamination sources and seasonal variations.

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